

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Design and Analysis of Algorithms**

Code No: **CSC325** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: complexity of the recurrence relation $T(n) = 2T(n/2) + 1$ is $O(n)$ using substitution method. 2. Write down advantages of dynamic programming over greedy strategy. Find optimal bracketing to multiply 4 matrices of order {2, 3, 4, 2, 5}. 3. Discuss heapify operation with example. Write down its algorithm and analyze its time and space complexity. Section B Attempt any Eight questions [8×5=40] 4. Define RAM model. Write down iterative algorithm for finding factorial and provide its detailed analysis. 5. Write down algorithm of insertion sort and analyze its time and space complexity. 6. Write down minmax algorithm and analyze its complexity. 7. When greedy strategy provides optimal solution? Write down job sequencing with deadlines algorithm and analyze its complexity. 8. Suppose that a message contains alphabet frequencies as given below, and find Huffman codes for each alphabet. Symbol Frequency a 30 b 20 c 25 d 15 e 35 9. Does backtracking give multiple solution? Trace the subset sum algorithm for the set {3, 5, 2, 4, 1} and sum = 8. 10. Why extended Euclidean algorithm is used? Write down its algorithm and analyze its complexity. 11. Define NP-complete problems with examples. Give brief proof of the statement "SAT is NP-complete". 12. Write short Notes on a) Aggregate Analysis b) Selection problems

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What is recurrence relation? How it can be solved? Show that time complexity of the recurrence relation $T(n) = 2T(n/2) + 1$ is $O(n)$ using substitution method.
2. Write down advantages of dynamic programming over greedy strategy. Find optimal bracketing to multiply 4 matrices of order {2, 3, 4, 2, 5}.
3. Discuss heapify operation with example. Write down its algorithm and analyze its time and space complexity.

Section B

Attempt any Eight questions [8×5=40]

4. Define RAM model. Write down iterative algorithm for finding factorial and provide its detailed analysis.
5. Write down algorithm of insertion sort and analyze its time and space complexity.
6. Write down minmax algorithm and analyze its complexity.
7. When greedy strategy provides optimal solution? Write down job sequencing with deadlines algorithm and analyze its complexity.
8. Suppose that a message contains alphabet frequencies as given below, and find Huffman codes for each alphabet. Symbol Frequency a 30 b 20 c 25 d 15 e 35
9. Does backtracking give multiple solution? Trace the subset sum algorithm for the set {3, 5, 2, 4, 1} and sum = 8.

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- 10.** Why extended Euclidean algorithm is used? Write down its algorithm and analyze its complexity.
 - 11.** Define NP-complete problems with examples. Give brief proof of the statement "SAT is NP-complete".
 - 12.** Write short Notes on a) Aggregate Analysis b) Selection problems